

WARNING TIME OPTIMISATION OF LOW-THRUST SATELLITE COLLISION AVOIDANCE USING DIFFERENTIAL ALGEBRA TECHNIQUES

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Increasingly, collision avoidance manoeuvres (CAMs) are needed in the operational lifetime of satellites. This work presents a novel methodology to optimise low-thrust CAMs with differential algebra techniques, offering a balance in accuracy and computational efficiency particularly suited for on-board CAM optimisation. In particular, this work keeps track of the time of maximum danger in a satellite conjunction with varying control, and leverages this tracking to find the latest possible time at which a thrust arc can be initiated while safely avoiding a collision between an operational satellite and a non-cooperative object.

INTRODUCTION

Recently, the increasing presence of space debris in Earth-centered orbits has posed significant challenges for satellite operators. As space debris objects orbit the Earth uncontrollably, they can pose a threat of collision to actively operated satellites nearby. Due to the high relative speeds involved, any collision would cause significant damage to satellites. Satellites are therefore forced to perform collision avoidance manoeuvres (CAMs) to stay clear of a collision when a close encounter is predicted to occur.

Many satellites in crowded operational orbits primarily use low-thrust propulsion systems, such as ion engines, Hall Effect thrusters and electrospray propulsion. Although these systems are more efficient than high-thrust propulsion, they reduce manoeuvrability, and manoeuvres such as CAMs can require minutes or even hours of thrust arcs. This limitation makes it important to start a satellite thrust arc adequately in advance, and can make optimisation of these arcs a complex task.

In literature, optimisation of CAM thrust arcs in terms of fuel consumption and Δv is often seen, which commonly leads to short thrust arcs long before a possible collision due to the natural orbital drift these burns provide. Operationally however, rather than thrusting as early as possible, CAMs are often performed as late as possible. The danger of a satellite-debris conjunction is often assessed in terms of the probability of collision (PoC), a metric based on the propagated position uncertainty distribution. This uncertainty is regularly updated through ongoing tracking of the debris object. More recent tracking data results in less positional uncertainty and therefore a more accurate assessment of conjunction danger and whether a CAM is needed or not.

Therefore, this work focuses on optimising the warning time for the CAM, that is, investigating the latest possible moment in time at which a thrust arc can be initiated to safely avoid a collision

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(i.e., decreasing the PoC to a safe threshold). Not only is this kind of optimisation pertinent to collision alerts which are received very late, but it can also be used to investigate whether there is enough time left to wait for the next batch of debris tracking data to arrive and re-assess whether a CAM is necessary. Earlier, this problem has been investigated from an optimal control standpoint.¹ While rigorous, it can be complicated to include various operational constraints in this methodology

Existing techniques for low-thrust CAM optimisation often rely on numerical integration, which can be computationally intensive, or analytical approximations, which may sacrifice accuracy. For example, Pavanello et al. explored recursive polynomial methods for designing fast low-thrust collision avoidance manoeuvres, emphasizing the potential for continuous optimization of thrust profiles.² Mahfouz et al. investigated fuel-optimal formation reconfiguration using unidirectional low-thrust propulsion systems, focusing on minimizing fuel consumption through advanced trajectory optimization techniques.³ Paduraru et al. applied reinforcement learning to optimize low-thrust CAMs, demonstrating improvements in computational efficiency compared to traditional methods.⁴

Despite these advances, limitations such as the high computational cost of numerical methods and the potential inaccuracies of analytical approximations persist. Differential Algebra (DA) offers a promising alternative by enabling efficient and accurate modeling of uncertainties and system dynamics. DA techniques facilitate real-time optimization, crucial for collision avoidance in dynamic and uncertain orbital environments, and are flexible regarding the inclusion of operational constraints. This work leverages DA to address these challenges, aiming to enhance both the precision and computational feasibility of low-thrust CAM optimisation.

Pavanello et al. have previously investigated low-thrust CAM optimisation, enabling efficient computation of the uncertainty distribution of objects involved in a conjunction at any point in time. Their approach integrates seamlessly with advanced higher-order methods, such as second-order cone programming, to address the non-convex nature of CAM optimisation problems.⁵ Inspired by their research, this work introduces a novel application of Differential Algebra (DA) by treating the time of maximum danger as a variable rather than a fixed parameter, as is commonly done. By allowing the time of maximum danger to vary, this methodology provides greater flexibility and accuracy and enables manoeuvres to shift the time of maximum danger further into the future.

In recent years, more attention has been paid to actively removing dangerous debris objects from orbit, but this work is still in its first steps. With more and more active satellites in orbit and enhanced debris tracking capabilities increasing the number of debris objects tracked, the number of CAMs required for safe space operations will only increase in coming years. Industry is working towards automated on-board collision avoidance, for which robust and efficient CAM optimisation algorithms are required. The DA methodology introduced in this work can contribute significantly to this goal.

This work is organised into a number of sections. First, the dynamics of the system and the close approach are presented, with the second part applying the DA techniques to the problem. Results about the performance of the methodology are then presented, as well as some strengths and weaknesses of the approach, followed by a conclusion.

PROBLEM DYNAMICS

We consider a controlled primary satellite described by a state

$$x_p = \begin{bmatrix} r_p \\ v_p \end{bmatrix} \in \mathbb{R}^6, \quad r_p \in \mathbb{R}^3 \text{ its position, and } v_p \in \mathbb{R}^3 \text{ its velocity,}$$

having a close encounter with a secondary uncontrolled satellite

$$x_s = \begin{bmatrix} r_s \\ v_s \end{bmatrix} \in \mathbb{R}^6, \quad r_s \in \mathbb{R}^3, \quad v_s \in \mathbb{R}^3.$$

The dynamics of the two objects are given by:

$$\frac{dx_p}{dt} = F(x_p) + \epsilon G(x_p)u, \quad (1)$$

$$\frac{dx_s}{dt} = F(x_s), \quad (2)$$

where $F(\cdot) : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ in Eq. (1) denotes the drift, representing uncontrolled motion of the objects, $\epsilon \in \mathbb{R}$ the maximum acceleration due to thrust, $G(x_p) : \mathbb{R}^6 \rightarrow \mathbb{R}^{6 \times 3}$ the influence of control on the dynamics, and $u \in \mathcal{U} \subset \mathbb{R}^3$ the normalized control, representing thrust acceleration. $F(\cdot)$ can include any dynamics, though in this work we limit the drift to two-body acceleration only.

We are primarily interested at the moment in time where the danger assessed by the danger metric is maximum, denoted by t_{MD} . In this work, we consider only *short encounters*, meaning that the danger metric is only assessed at t_{MD} , rather than in an entire time interval. It is therefore sufficient to design a collision avoidance manoeuvre which changes the danger metric to safety at t_{MD} only.

To evaluate the safety of a close approach, three commonly used metrics are often considered. The first is the Euclidean distance, defined as

$$d_E = \|r_p - r_s\|, \quad (3)$$

where r_p and r_s are the positions of the primary and secondary satellites, respectively, and $\|\cdot\|$ denotes the Euclidean norm.

The second metric is the probability-of-collision formula, which estimates the likelihood of collision based on the relative position and positional uncertainty. For two satellites with Gaussian-distributed positional uncertainties, the probability of collision can be expressed as

$$d_{PoC} = \frac{R^2}{2\pi\sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2}(r_p - r_s)^\top \Sigma^{-1}(r_p - r_s)\right), \quad (4)$$

where $\Sigma = \Sigma_p + \Sigma_s$ is the combined covariance matrix, with Σ_p and Σ_s representing the covariance matrices of the primary and secondary satellites, respectively. R represents the hard-body radius of the two objects in the conjunction.

The third commonly used metric is the Mahalanobis distance, which accounts for the shape and orientation of the positional uncertainty ellipsoids and is defined as

$$d_M = \sqrt{(r_p - r_s)^\top \Sigma^{-1}(r_p - r_s)}. \quad (5)$$

The Mahalanobis distance provides a normalised measure of separation that incorporates both the relative distance and the positional uncertainties. These metrics offer complementary perspectives on collision risk and are used depending on the context and precision required.

From now on, $d_{(\cdot)}$ represents the danger of distance metric of a conjunction, without specifying the exact metric choice.

Optimisation Problem

Regardless of the danger metric used, the objective of a Collision Avoidance Maneuver (CAM) is to reduce this metric to a safe threshold at t_{MD} , thereby ensuring that the threshold is not exceeded for any t during short encounters. The corresponding optimisation problem is formulated as follows:

$$\begin{aligned}
& \min_{t_s \leq 0} -t_s \quad \text{subject to:} \\
& \frac{dx_p}{dt} = F(x) + \varepsilon G(x) u, \quad \forall t \in [t_0, t_{MD}], \\
& 0 \leq u(t) \leq 1, \quad u \in \mathbb{R}, \quad \forall t \in [t_0, t_{MD}], \\
& x_p(t_s) = x_{p,t_s}, \\
& d_{(\cdot)}(t_f) = \sigma_{(\cdot)}, \\
& t_f = t_{MD}(t_f),
\end{aligned} \tag{6}$$

where t_s and t_f denote the start and end times of the problem (and the CAM thrust arc), respectively. Here, t_{MD} is nominally assumed to be zero, meaning that t_s is negative, and minimising $-t_s$ corresponds to maximising the thrust initiation time. The initial state $x_p(t_s)$ is determined by propagating the natural drift of the primary satellite backward in time from the current state. The parameter σ represents the safety threshold, which depends on the specific danger metric employed. Operationally, values like $\sigma_E = 1$ km as a minimum Euclidean distance or $\sigma_{PoC} = 10^{-4}$ as a maximum PoC are often used.⁶ Finally, the condition $t_f = t_{MD}(t_f)$ ensures that the end time of the thrust arc and the optimisation problem coincides with the moment of maximum danger; the thrust arc is then ensured to be contained to the time interval $[t_s, t_{MD}]$.

Since t_{MD} can vary depending on the control input u , it is necessary to track its evolution as the control changes. By approximation, we leverage a property of the Euclidean distance metric, which states that:

$$(r_p(t_{MD}) - r_s(t_{MD})) \cdot (v_p(t_{MD}) - v_s(t_{MD})) = 0, \tag{7}$$

where the relative position vector and the relative velocity vector are orthogonal at the moment of geometric closest approach. While formulating the time of maximum danger for the Probability of Collision (PoC) and Mahalanobis distance metrics is less straightforward, for the purposes of this work, the relationship in Eq. (7) is used to approximate t_{MD} . While non-Euclidean metrics such as PoC and Mahalanobis distance consider additional factors like uncertainty distributions, the geometric closest approach often aligns closely with the time of maximum danger. For small uncertainties, these metrics are dominated by the relative distance, validating the use of orthogonality as a first-order approximation.

DIFFERENTIAL ALGEBRA APPROACH

In order to use differential algebra effectively in the context of the optimisation problem shown in Eq. (6), it is convenient to discretise the time grid between t^{alert} (the time at which a dangerous conjunction in the future is noticed) and $\bar{t}_{MD}$ (the nominal time of maximum danger), where $t^{alert} < t_s < t_{MD}$. Obtained is a discretised time grid $t_0, t_1, \dots, t_N$ of N intervals, where $t_0 = t^{alert}$ and $t_N = \bar{t}_{MD}$.

We consider $r_p(t_{N-1})$ and $v_p(t_{N-1})$, the state of the primary satellite at time t_{N-1} , as:

$$r_p(t_{N-1}) = \bar{r}_p(t_{N-1}) + \delta r_p(t_{N-1}), \quad (8)$$

$$v_p(t_{N-1}) = \bar{v}_p(t_{N-1}) + \delta v_p(t_{N-1}), \quad (9)$$

that is to say, they are vectors described by the sum of their nominal state ($\bar{r}_p(t_{N-1})$ and $\bar{v}_p(t_{N-1})$) and possible perturbations ($\delta r_p(t_{N-1})$ and $\delta v_p(t_{N-1})$) caused by control u . The state of the secondary object is unchanged:

$$r_s(t_{N-1}) = \bar{r}_s(t_{N-1}), \quad v_s(t_{N-1}) = \bar{v}_s(t_{N-1}),$$

as the control u does not act on the secondary object.

We consider a possible thrust duration from $t_0 = t^{alert}$ to $t_N = \bar{t}_{MD}$, assuming that the optimal time of starting thrust t_s is within this range. We introduce a change of variable:

$$\tau = \frac{t}{T} \quad (10)$$

, where $T = \bar{t}_{MD} - t_0$, leading to $\frac{d}{dt} = \frac{d}{d\tau} \frac{d\tau}{dt} = \frac{d}{d\tau} \frac{1}{T}$. Note that T can vary under the influence of control, as the time of maximum danger t_{MD} is strongly dependent on the orbit of the primary satellite. We can therefore write

$$T(t_{N-1}) = \bar{t}_{MD} + \delta t = t_0 \quad (11)$$

for some perturbation δt . Lastly, we assume a constant thrust profile $u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t) = 0 + \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t)$ in RSW components in the time interval from t_{N-1} to $\bar{t}_{MD} + \delta t$: a zero nominal thrust profile and a thrust perturbation δu .

Consider now the mapping of x_p at $t = t_{N-1}$ to $t = \bar{t}_{MD} + \delta t$, for example through a high-fidelity numerical propagator. Equation Eq. (12) maps the perturbed state of the primary satellite at time t_{N-1} to the nominal time of maximum danger $\bar{t}_{MD}$, incorporating deviations caused by control actions. This allows us to express the system's behavior as Taylor polynomials in terms of perturbations δr , δv , δu and δt .

$$\begin{cases} r_p(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t), \delta t) \\ v_p(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t), \delta t) \\ r_s(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta t) \\ v_s(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta t) \\ \Delta r(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t), \delta t) \\ \Delta v(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t), \delta t) \\ \Delta \tau(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t), \delta t) \end{cases} \quad (12)$$

Here, $\Delta r = r_p - r_s$ and $\Delta v = v_p - v_s$ are defined as the relative position and velocity respectively, in the case of Eq. (12) at time $\bar{t}_{MD} + \delta t$, and $\Delta \tau$ the change in thrust duration as a result of the aforementioned perturbations. Note that δr and δv cover perturbations at t_{N-1} , δu covers a perturbation in the interval $[t_{N-1}, \bar{t}_{MD} + \delta t]$ and δt covers a perturbation at the end of the time interval $[t_{N-1}, \bar{t}_{MD}]$. The state of the secondary satellite is unaffected by control, and the Taylor polynomials describing the state in Eq. (12) are therefore only dependent on δt .

To ensure the control-induced perturbations accurately mitigate collision risk, δt must satisfy the closest approach condition, where the relative position and velocity vectors are orthogonal. Equation Eq. (13) derives this δt through polynomial inversion, linking perturbations at t_{N-1} to the geometric closest approach.

$$\delta t = \delta t_{CA} = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{CA} + \delta t)) \quad (13)$$

The orthogonality condition $\Delta r \cdot \Delta v = 0$ at t_{MD} ensures that the derived δt corresponds to the time when the relative distance between the objects is minimal. This guarantees the perturbations are accurately aligned with the dynamics of the conjunction. Resolving δt ensures that all evaluations at $\bar{t}_{MD} + \delta t$ track the moment of maximum danger.

Next, the danger metric at $\bar{t}_{MD} + \delta t$ (tracking t_{MD}) can be approximated by a Taylor polynomial like in Eq. (12). Note that all danger metrics described in Eq. (3), Eq. (4) and Eq. (5) are primarily dependent on Δr . We therefore also have

$$d_{(\cdot)}(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t), \delta t) \quad (14)$$

We consider the expansion of $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$ in u :

$$d_{(\cdot)}(\bar{t}_{MD} + \delta t) = d_{(\cdot)}(\bar{t}_{MD}) + \frac{\partial d_{(\cdot)}}{\partial \delta u} \delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t) + \mathcal{O}(\|\delta u\|^2) \quad (15)$$

In low-thrust propulsion scenarios, the limited thrust magnitude ($\epsilon \ll 1$) necessitates precise targeting of the danger metric's rate of change to achieve optimal collision avoidance. By maximising the gradient of the danger metric with respect to the control vector δu , the algorithm ensures that the control action is directed for maximum effectiveness:

$$\delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t) = \epsilon \frac{\frac{\partial d_{(\cdot)}}{\partial \delta u}}{\|\frac{\partial d_{(\cdot)}}{\partial \delta u}\|} \quad (16)$$

Eq. (16) can be substituted in Eq. (14), together with $\delta r(t_{N-1}) = \delta v(t_{N-1}) = 0$ to investigate the effect of $\delta u(t_{N-1} \rightarrow \bar{t}_{MD} + \delta t)$ on $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$. There are now two cases possible:

- $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$ has exceeded the safety threshold $\sigma_{(\cdot)}$, and $t_{N-1} < t_s < t_N$. A refinement in the interval $[t_{N-1}, t_N]$ can be performed to obtain t_s with more precision
- $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$ has not yet reached the safety threshold $\sigma_{(\cdot)}$. In that case, we consider the time interval $[t_{N-2}, t_{N-1}]$ and obtain the following system:

$$\begin{cases} r_p(t_{N-1}) = \mathcal{T}(\delta r(t_{N-2}), \delta v(t_{N-2}), \delta u(t_{N-2} \rightarrow t_{N-1}), \delta t) \\ v_p(t_{N-1}) = \mathcal{T}(\delta r(t_{N-2}), \delta v(t_{N-2}), \delta u(t_{N-2} \rightarrow t_{N-1}), \delta t) \\ r_s(t_{N-1}) = \mathcal{T}(\delta t) \\ v_s(t_{N-1}) = \mathcal{T}(\delta t) \end{cases} \quad (17)$$

Polynomial inversion can be used to obtain δt again, ensuring a tracking of the closest approach, and $r_p(t_{N-1})$, $v_p(t_{N-1})$, $r_s(t_{N-1})$ and $v_s(t_{N-1})$ can be substituted in Eq. (14), together with Eq. (16), obtaining $\delta u(t_{N-2} \rightarrow t_{N-1})$. The same process can now be repeated.

A backward sweep is therefore performed over the time intervals $[t_{N-1}, t_N + \delta t]$, $[t_{N-2}, t_{N-1}]$, ..., $[t_0, t_1]$ until the safety threshold $\sigma(\cdot)$ is reached, depending on the exact danger metric, and t_s can be found accordingly.

APPLICATIONS

While the proposed methodology is valid for a wide range of satellite conjunctions, the geometry of the conjunction can play a significant role, not only in the control profile $u(t)$, but also in the methodology accuracy. As mentioned, the control profile requires discretisation in the DA methodology described in this work and is assumed to have constant components in the time intervals. Therefore, when the optimal control profile changes rapidly, the discretised control might be less accurate. Additionally, the combined covariance of the primary and secondary objects changes over time for non-circular orbits, increasing near the periapsis and decreasing near the apsis.

In this section, the algorithm is applied to one/two representative test cases:

- An ISS-like orbit with zero eccentricity in Low-Earth Orbit
- (Possibly?) A GTO-like eccentric orbit with high eccentricity and altitude ranging from 500 to 40000 km

Details of the geometries and the case studies are shown in Table 1.

In the Results section, the control profiles $u(t)$ are investigated for the two case studies and the different danger metrics d_E , d_{PoC} and d_M . The number of nodes N and the true anomaly at $\bar{t}_{DM}$ are varied to highlight their influence.

RESULTS

The discretised control profile $u(t)$ of the first case study with an ISS-like orbit is shown in Figure 1, for three different distance metrics (FIGURE NEEDS BETTER LABELLING AND COLOURING):

The discrete nature of the control in Figure 1 is clear, as the control components are constant within the discrete time intervals. The Mahalanobis and PoC-based control profiles are nearly identical, as both depend heavily on the covariance matrix Σ . They differ slightly from the control profile based on the Euclidean distance danger metric, but not significantly. Vertical lines in Figure 1 indicate the maximum t_s values for each distance metric. These values are not the same for the three danger metrics, as d_M and d_{PoC} depend to a large extent on the combined covariance Σ , while d_E does not.

Additionally, the change of the danger metrics throughout the backward sweep is shown in Figure 2 (NB CURRENTLY ONLY EUCLIDEAN SHOWN)

The thrust duration is shown in Figure 2, rather than the time t itself, coinciding with $t_{DM} - t_s$. t_{DM} varies with time in principle, but the change from its nominal value is less than a second and not visible in Figure 1 and Figure 2. Since the control is unconstrained in direction, the Euclidean distance metric is continuously increasing with thrust until the safe threshold is reached.

Table 1. Parameters of the Case Studies

Parameter	Case Study 1 (ISS-like)	Case Study 2 (GTO-like)
Semi-major axis, a (km)	6780	24582
Eccentricity, e	0	0.73
Inclination, i (deg)	51.6	27
RAAN, Ω (deg)	0	0
Argument of Perigee, ω (deg)	0	270
True Anomaly, ν at $\bar{t}_{DM}$ (deg)	30	90
Thrust Magnitude, ϵ (m/s ²)	$1.0 \cdot 10^{-4}$	
Gravitational Parameter, μ (m ³ /s ²)	$3.986 \cdot 10^{14}$	
Number of Nodes, N	50	
Alert Time, t^{alert} (s)	-1800	
Nominal miss distance, d_E (m)	200	
Safe miss distance, σ_E (m)	1000	
Nominal covariance, Σ_{xx} ()		
Nominal covariance, Σ_{xy} ()		
Nominal covariance, Σ_{yy} ()		
Safe PoC, σ_{PoC}		
Safe Mahalanobis distance, σ_M ()		

CONCLUSION AND DISCUSSION

The introduced DA methodology provides several advantages for low-thrust CAM optimisation. First, the control profile minimizing $-t_s$ is quickly determined using a first-order approximation. Given the small thrust magnitudes involved, this first-order solution is often sufficiently accurate. If needed, it can be refined to second-order accuracy by extending the Taylor expansion of the danger metric $d_{(\cdot)}$. Furthermore, the methodology is flexible, allowing the use of various danger metrics beyond the three mentioned in this work. The combined covariance matrix Σ can also be expanded as a Taylor polynomial to enhance accuracy, if necessary. Additionally, operational constraints, such as restricted thrust intervals or directional limitations, can be seamlessly integrated into the DA framework.

One of the methodology's key strengths is its explicit tracking of t_{MD} , the time of maximum danger. This is achieved through polynomial inversion of the δt term, not only in the time interval $[t_{N-1}, \bar{t}_{MD} + \delta t]$ but also in earlier intervals. During the backward sweep, t_{MD} is accurately updated through the Taylor polynomial representing $\Delta r(\bar{t}_{MD} + \delta t)$, ensuring the danger metric $d_{(\cdot)}$ is evaluated at the correct time. However, the control profile $u_{t_{N-1} \rightarrow \bar{t}_{MD} + \delta t}$ is not recalculated in earlier intervals. In other words, once δt is resolved for $[t_{N-1}, \bar{t}_{MD} + \delta t]$, the thrust continues

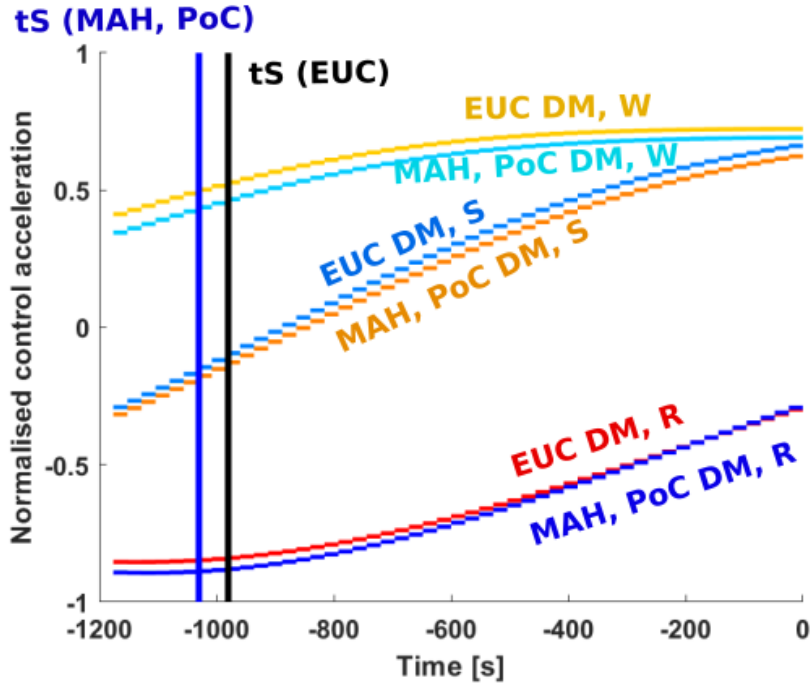


Figure 1. The control profile over time for the Euclidean, Mahalanobis and PoC distance metrics, and the latest time to initiate thrust

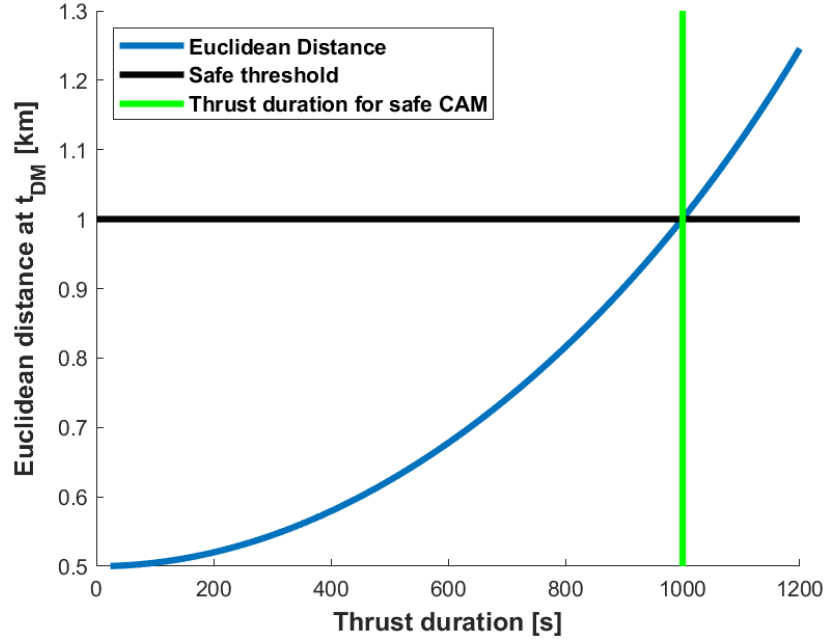


Figure 2. The Euclidean distance metric for various thrust durations

uninterrupted until the recalculated t_{MD} . Earlier control profiles $u_{t_{i-1} \rightarrow t_i}$ ($i = 1, 2, \dots, N - 1$) adjust t_{MD} as δt evolves, but they do not retroactively modify the control in subsequent intervals.

Despite these advantages, the methodology has some limitations. The control profile u must be discretized over time, with constant components in each time interval. Choosing an appropriate discretization number N can be challenging, and inaccuracies may arise if the true optimal control profile u^* (found e.g. using optimal control theory) varies rapidly within an interval. Additionally, the methodology heavily depends on two key assumptions: (1) that the time of maximum danger coincides with the geometric closest approach, and (2) that thrust magnitudes remain small relative to the natural dynamics. As a result, this approach is unsuitable for impulsive CAMs or long encounters with small $\|\Delta v\|$ values between the primary and secondary.

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